

XAI for Forest Fire Risk Assessment

Visualizing Meteorological Drivers with SHAP
Frameworks

University of Trieste | Data Visualization
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01. RESEARCH CONTEXT

Dataset Summary

Origin: Montesinho Natural Park (Portugal).

Telemetry: Real-time weather data (Temp, RH, Wind, Rain).

Objective: Predict burned area using Random Forest Regressors.

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Analytical Challenge

Machine Learning models are typically "black-boxes."

By applying **SHAP (SHapley Additive exPlanations)**, we map mathematical weights into intuitive **Visual Channels** for human interpretation.

02. ADDRESSED QUESTIONS

1. How do individual meteorological factors globally impact the prediction of fire risk?

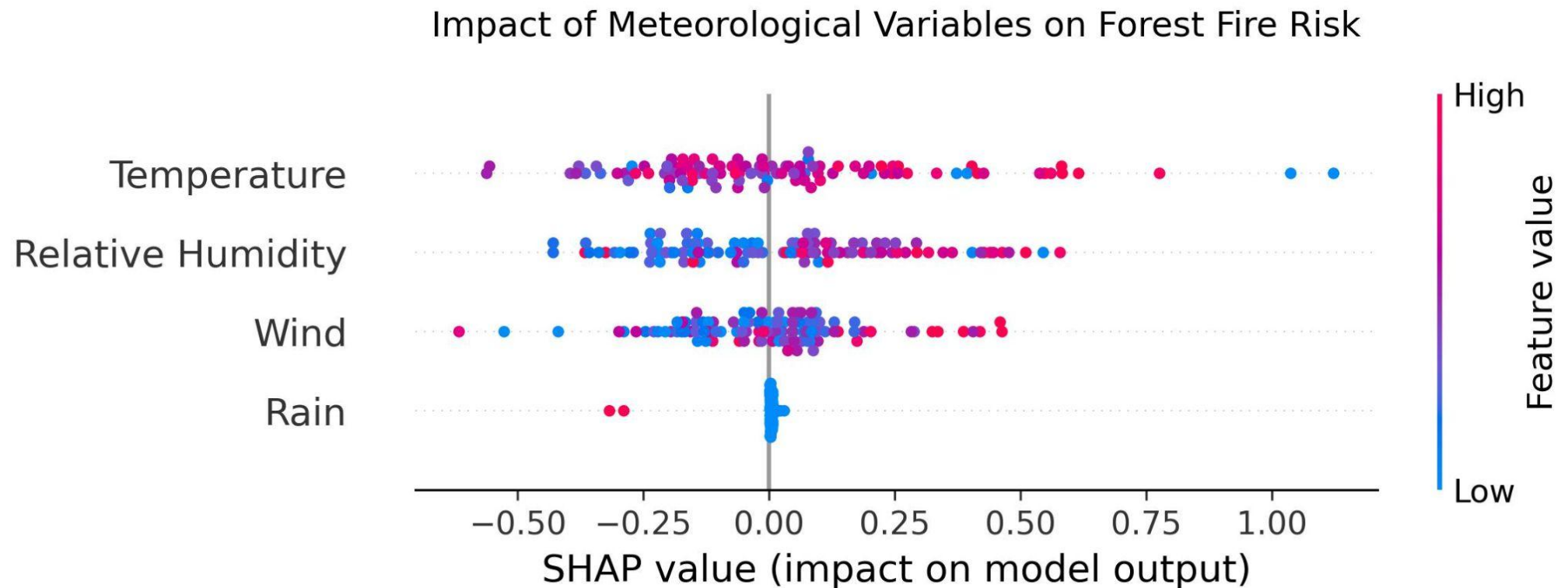
02. ADDRESSED QUESTIONS

1. How do individual meteorological factors globally impact the prediction of fire risk?

2. What specific weather triggers drove the prediction for a high-risk case study (Day 5)?

03. GLOBAL IMPACT ANALYSIS

The model captures counterintuitive relationships: higher temperatures unexpectedly lower predicted risk, exposing complex variable interactions.



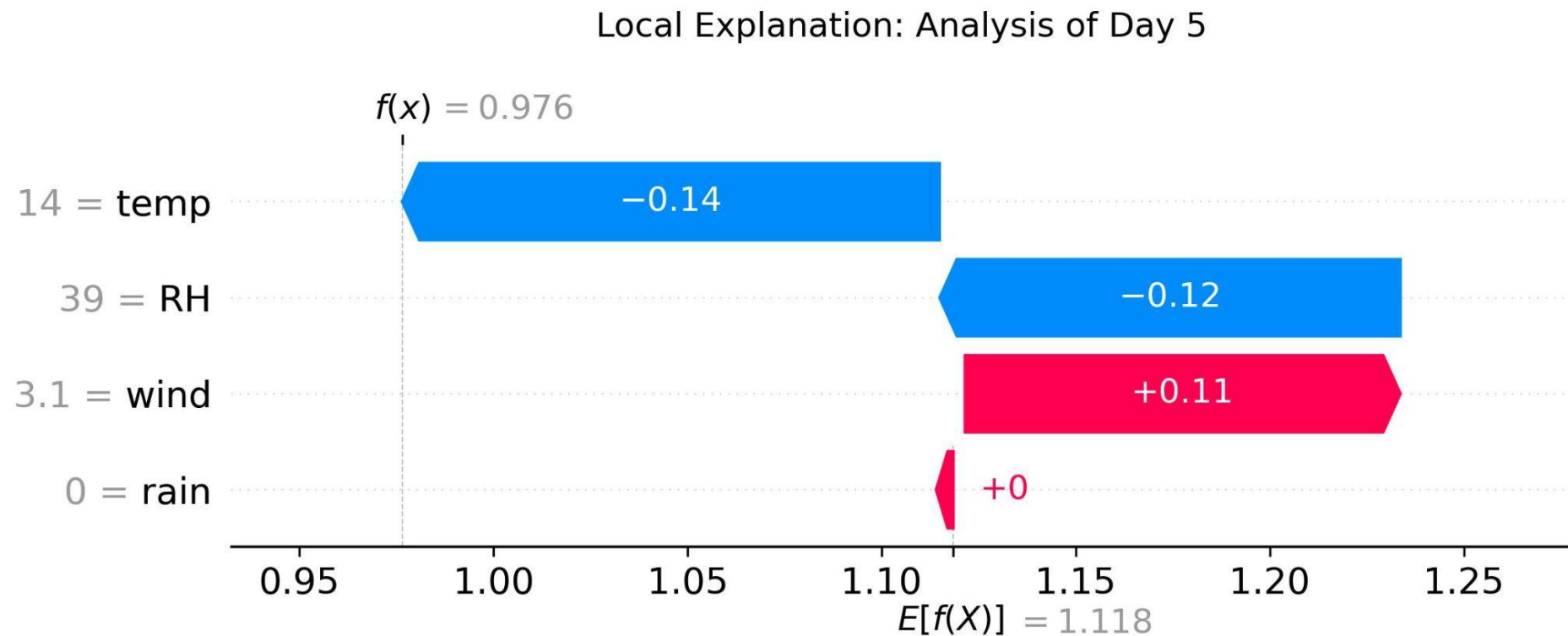
03. MAPPING: GLOBAL VISUALIZATION

Data Attribute	Attribute Type	Visual Channel Assigned
Feature Importance	Nominal / Ranked	Vertical Position (Y-axis Order)
Impact on Prediction	Quantitative	Horizontal Position (X-axis)
Predictor Value	Quantitative	Color (Diverging Blue-to-Red)
Sample Density	Quantitative	Vertical Jitter (Point Clusters)

Note the **non-linear interactions**: even at low temperatures (blue dots), extreme wind or drought conditions can push risk to the right.

04. LOCAL EVENT BREAKDOWN

On Day 5, the cooling effects of temperature (14) and humidity (39) outweighed wind impact (+0.11), dropping the final risk prediction to 0.976.



04. MAPPING: LOCAL VISUALIZATION

Data Attribute	Attribute Type	Visual Channel Assigned
Expected Baseline	Quantitative	X-Axis Origin Point
Feature Contribution	Quantitative	Bar Horizontal Width (Size)
Impact Direction	Categorical	Color (Pink: Increase / Blue: Decrease)
Final Output Area	Quantitative	Final X-Axis Endpoint

This sequential decomposition (Waterfall) minimizes cognitive load while identifying specific environmental triggers for a single prediction.

Conclusion

Black-box models can inherit historical data biases, learning **false physical relationships**. **XAI** visualizations act as a critical audit tool, exposing these **flaws to prevent blind trust** in automated systems
